

LÉO MOULIER

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M2 STAPS STUDENT |SPORTS ENGINEERING, DATA SCIENCE & PERFORMANCE

PROFESSIONAL SUMMARY

Master's student in Sports Science (STAPS) with coursework experience in multi-view human motion capture, pose estimation and machine learning. Hands-on with Pose2Sim, camera calibration, triangulation, trajectory filtering and Python/MATLAB analysis across boxing, parkour, fitness and basketball projects. Interested in robust 3D human reconstruction for close-contact sports; available February–June 2027.

RELEVANT PROJECTS

- **Multi-camera markerless motion capture (coursework)**

Pose2Sim, OpenPose, MediaPipe and YOLO-Pose; multiple-smartphone video synchronization/management, intrinsic/extrinsic calibration, epipolar geometry and triangulation. Tracked multiple athletes, including re-identification; low-pass filtered trajectories. Worked through occlusion and missing-marker challenges in sports motion capture (boxing, parkour, fitness, basketball).

- **Machine learning on handwritten digits**

Used scikit-learn and OpenCV for classification, regression and detection; explored Random Forest, Decision Tree and scikit-learn MLP models, compared accuracy and optimized models.

- **Boxing VR reflex-training game (Unity)**

Marker-based capture with three infrared and three standard cameras at 100 Hz. Cleaned captures in acquisition software; recorded one performer's eight punches, four dodges and 12 sequences. Integrated motion in Unity; used Blender/OpenSim. Delivered a functional game to train/test dodge reflexes.

- **Biomechanics & scientific writing**

OpenSim/Mokka analysis of a parkour cat leap (table height and jump length). Reports/articles in master's-thesis style on "galvanic vestibular stimulation for simulated free-fall in VR" and "IMU vs Zeno gait analysis"; literature review. Articles in progress.

SKILLS

Computer vision & motion capture: OpenPose, MediaPipe, YOLO-Pose; Pose2Sim; multi-person tracking/re-identification; multi-view calibration, synchronization, triangulation, epipolar geometry; marker-based capture; trajectory filtering.

Machine learning: scikit-learn (Random Forest, Decision Tree, MLP), OpenCV; classification, regression, detection; model comparison/optimization.

3D Modelling: FreeCAD/SolidWorks, Blender.

Programming & tools: Python (NumPy, pandas, matplotlib; 2 years in Master's data analysis, ML and 3D visualization), MATLAB, OpenSim, Mokka, Unity, Blender, Conda, GitHub.

Areas of Interest: VR/XR/AR, electronics, Web3, marketing, prototyping, human–machine interfaces, design, nutrition, AI.

Sports: Climbing, MMA, skiing (All-mountain/Freestyle), football, DH MTB, weight training, volleyball, canyoning, gymnastics.

Languages: French (**native**); English (**B2/C1**, Cambridge certificate); German (**B1**).

EDUCATION

Université de Reims Champagne Ardenne — *Reims | September 2025 - Present*
Master's Degree in Sports Science (STAPS) : Physical Activity Engineering and Ergonomics; Speciality in Innovation, Design, Data Science and Performance

Université Jean Monnet — *Saint-Etienne | September 2021 - July 2024*
Bachelor's Degree in Sports Science (STAPS) *: Year 3: Physical Education Year 2: Sports Training*

Lycée Albert Camus — *Firminy | September 2018 – July 2021*
French Baccalaureate : Specialisations: Life and Earth Sciences, Physics and Chemistry; additional Mathematics

ADDITIONAL EXPERIENCE

Stunt Performer — Freelance / Project-based
Film productions, short films, and music videos — *France | June 2023 - Present*

Motion Model
rerved as motion model in three workshops: parkour cat leap, boxer punches/dodges and gymnast backflip. — *Reims | September 2025 - October 2026*

Earlier roles:
climbing instructor (Club Cimes & Rocs, 2019–2021);
physical education teaching internship (Jan–Apr 2024);
retail, **food service**, **postal** and **adventure-park** work.

CERTIFICATIONS

- **MATLAB Onramp, MathWorks** — *October 2025*
- **Cambridge English Certificate, Cambridge Assessement English** — *March 2021*